Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 1.1 · Getting to grips with indices

Consider the sequence (u_n) defined for every natural number n by $u_n = 2n^2 - 3n$. The first term is therefore the one of index 0.

1. Calculate u_0, u_1 and u_4 separately, showing the substitution of n.
2. Express u_{n+1} in terms of n, then expand and simplify this expression.
3. Calculate $u_{n+1} - u_n$. By studying its sign for integer indices, describe the variations of the sequence.

Exercise 1.2 · Choosing the right model

A nursery currently has 100 young plants. It is considering two models to forecast its stock: in model A, the stock increases by 15 plants per month; in model B, it increases by 10 percent per month. Let a_n and b_n be the forecast stocks after n months, with $a_0 = b_0 = 100$. The two models are studied separately.

1. **Model A.** Write the relation between a_{n+1} and a_n. Recognise the type of sequence, then express a_n in terms of n. Calculate a_2.
2. **Model B, after studying geometric sequences.** Answer the same questions for (b_n) and calculate b_2.
3. For each model, explain what is assumed to be constant. Give one reason why a very long-term forecast might become unrealistic.

Exercise 1.3 · Recovering a sequence

The sequence (u_n) is geometric, defined from index 0. Its common ratio q is strictly positive. We only know $u_1 = 6$ and $u_3 = 54$.

1. Determine q by relating the two known terms. Present all the steps that lead to choosing its value.
2. Deduce u_0, then u_5.
3. What would change if the hypothesis $q > 0$ were removed? Give all the possible reasons and explain whether the value of u_5 would be affected.

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Exercise 1.4 · Rows of seats

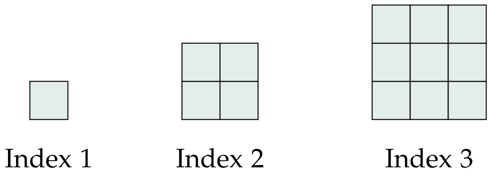
A theatre has 12 rows. The first row has 18 seats; each following row has two more seats than the previous one. Let u_n be the number of seats in row n, for $1 \leq n \leq 12$.

1. Define the sequence (u_n) and calculate the number of seats in the last row.
2. Calculate the total capacity of the theatre. Explain the number of terms used in the sum.
3. Calculate the average number of seats per row and use this result to check the total found.

Exercise 1.5 · Accumulating squares

Patterns are built with identical square tiles. The pattern of index 1 contains one tile. To go from the pattern of index n to that of index $n+1$, we add $2n+1$ tiles. Let u_n be the number of tiles in the pattern of index n, for $n \geq 1$.

1. Calculate u_2 and u_3. Draw an arrangement of these tiles that suggests a formula for u_n.
2. Conjecture an expression for u_n in terms of n.
3. Check that this expression gives the first term and satisfies the construction rule. Explain why these two checks identify the sequence.



One possible arrangement of the first three patterns; the drawing helps to state a conjecture, which still has to be justified.

Exercise 1.6 · Two repayments

A person must repay 600 monetary units, with no interest. Two plans are considered. With plan A, they pay 50 units at the end of each month until the debt is repaid. With plan B, they pay each month half of what they still owe. The calculations for model B are carried out without rounding. Let a_n and b_n be the remaining debts after the payment of month n, with $a_0 = b_0 = 600$.

1. Express a_n until the debt is cleared. In which month is the repayment completed?
2. Express b_n. Determine the first month in which the remaining debt is strictly less than 10 units; check the previous month.
3. In model B without rounding, does the debt become exactly zero after a finite number of months? Justify and comment on the difference with a real situation.

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Exercise 1.7 · Reading a list

A program must keep the terms of the sequence defined by $u_0 = 2$ and $u_{n+1} = 3u_n - 1$. It starts with the instructions `u=2` and `L=[u]`. It then executes five times, in this order, `u=3*u-1` then `L.append(u)`.

1. Execute the first two passes of the loop by hand. Then give the contents of the list.
2. How many elements does the list contain at the end of the five passes? What is the index of the last term stored?
3. Write the complete program in Python, then check the value of the last element. Explain why the number of terms exceeds the number of passes by one.

Exercise 1.8 · Delayed review

This question is done one week after the chapter, without consulting the formulas. A sequence is defined for $n \geq 2$ by $u_2 = 12$ and $u_{n+1} = u_n - 3$.

1. Calculate u_7, stating how many passes separate the indices 2 and 7.
2. Calculate the sum $u_2 + u_3 + \cdots + u_7$. Explain the difference between the number of terms in this sum and the number of passes calculated in the first question.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 9 p. 64, questions a and b. Calculate the first four terms. Write the substitutions before simplifying.

CQFD, exercise 11 p. 64, all questions. Calculate the terms one after another and keep a record of the four steps.